

Группа: M3206

К работе допущен:

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Работа выполнена:

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Отчет принят:

## Рабочий протокол и отчет по лабораторной работе № 5

### Управление памятью в ОС Linux

#### Подготовка к проведению экспериментов

Зафиксируйте в отчете данные о текущей конфигурации операционной системы в аспекте управления памятью:

1. Общий объем оперативной памяти.
2. Объем раздела подкачки.
3. Размер страницы виртуальной памяти.
4. Объем свободной физической памяти в ненагруженной системе.
5. Объем свободного пространства в разделе подкачки в ненагруженной системе.

```
[root@localhost lab5]# free -h
```

|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 3.7Gi | 169Mi | 3.3Gi | 8.0Mi  | 218Mi      | 3.3Gi     |
| Swap: | 819Mi | 0B    | 819Mi |        |            |           |

```
[root@localhost lab5]# getconf PAGE_SIZE
```

4096

#### Эксперимент №1

1. Этап 1

```
#!/bin/bash

: > report.log

arr=()
stop=0

while true
do
    arr+=(1 2 3 4 5 6 7 8 9 10)
    ((step++))

    if ((step % 100000 == 0)); then
        echo "${arr[@]}" >> report.log
    fi
done
```

```
top - 23:44:44 up 2 min, 1 user, load average: 1.46, 0.80, 0.32
Tasks: 114 total, 3 running, 111 sleeping, 0 stopped, 0 zombie
%Cpu(s): 74.4 us, 24.0 sy, 0.0 ni, 0.0 id, 0.0 wa, 1.7 hi, 0.0 si, 0.0 st
MiB Mem : 3778.4 total, 2866.2 free, 687.0 used, 225.2 buff/cache
MiB Swap: 820.0 total, 820.0 free, 0.0 used, 2863.7 avail Mem
```

| PID  | USER | PR | NI | UIRT   | RES    | SHR   | S | %CPU | %MEM | TIME+   | COMMAND                            |
|------|------|----|----|--------|--------|-------|---|------|------|---------|------------------------------------|
| 1452 | root | 20 | 0  | 753788 | 534276 | 2840  | R | 96.1 | 13.8 | 0:37.98 | nen.bash                           |
| 1453 | root | 20 | 0  | 274288 | 4532   | 3864  | R | 1.3  | 0.1  | 0:00.54 | top                                |
| 372  | root | 20 | 0  | 0      | 0      | 0     | I | 0.3  | 0.0  | 0:00.54 | kworker/0:3-events_power_efficient |
| 875  | root | 20 | 0  | 430980 | 38084  | 36476 | S | 0.3  | 1.0  | 0:00.69 | sssd_nss                           |
| 903  | root | 20 | 0  | 235036 | 3624   | 3244  | S | 0.3  | 0.1  | 0:00.25 | bash                               |
| 1    | root | 20 | 0  | 179204 | 13384  | 8412  | S | 0.0  | 0.3  | 0:08.77 | systemd                            |
| 2    | root | 20 | 0  | 0      | 0      | 0     | S | 0.0  | 0.0  | 0:00.01 | kthreadd                           |

```
top - 23:51:33 up 9 min, 1 user, load average: 1.42, 1.15, 0.66
Tasks: 107 total, 4 running, 103 sleeping, 0 stopped, 0 zombie
%Cpu(s): 97.1 us, 1.2 sy, 0.0 ni, 0.0 id, 0.0 wa, 1.6 hi, 0.0 si, 0.0 st
MiB Mem : 3778.4 total, 959.1 free, 2505.9 used, 313.5 buff/cache
MiB Swap: 820.0 total, 820.0 free, 0.0 used, 1044.8 avail Mem
```

| PID  | USER | PR | NI | UIRT    | RES   | SHR  | S | %CPU | %MEM | TIME+   | COMMAND    |
|------|------|----|----|---------|-------|------|---|------|------|---------|------------|
| 1452 | root | 20 | 0  | 2613140 | 2.3g  | 2840 | R | 97.4 | 61.9 | 7:13.56 | nen.bash   |
| 1453 | root | 20 | 0  | 274288  | 4532  | 3864 | R | 1.0  | 0.1  | 0:05.16 | top        |
| 23   | root | 39 | 19 | 0       | 0     | 0    | R | 0.3  | 0.0  | 0:00.41 | khugepaged |
| 903  | root | 20 | 0  | 235036  | 3624  | 3244 | S | 0.3  | 0.1  | 0:00.66 | bash       |
| 1    | root | 20 | 0  | 179204  | 13384 | 8412 | S | 0.0  | 0.3  | 0:08.78 | systemd    |
| 2    | root | 20 | 0  | 0       | 0     | 0    | S | 0.0  | 0.0  | 0:00.02 | kthreadd   |

```
top - 23:55:20 up 12 min, 1 user, load average: 2.25, 1.60, 0.94
Tasks: 108 total, 5 running, 103 sleeping, 0 stopped, 0 zombie
%Cpu(s): 74.4 us, 23.1 sy, 0.9 ni, 0.0 id, 0.0 wa, 1.7 hi, 0.0 si, 0.0 st
MiB Mem : 3778.4 total, 114.0 free, 3304.1 used, 360.3 buff/cache
MiB Swap: 820.0 total, 819.7 free, 0.3 used, 246.1 avail Mem
```

| PID  | USER | PR | NI | UIRT    | RES   | SHR   | S | %CPU | %MEM | TIME+    | COMMAND    |
|------|------|----|----|---------|-------|-------|---|------|------|----------|------------|
| 1452 | root | 20 | 0  | 3410024 | 3.0g  | 2812  | R | 93.5 | 82.5 | 10:51.31 | mem.bash   |
| 1455 | root | 39 | 19 | 538556  | 41008 | 23452 | R | 1.6  | 1.1  | 0:01.80  | dnf        |
| 1453 | root | 20 | 0  | 274288  | 4488  | 3820  | R | 1.3  | 0.1  | 0:07.70  | top        |
| 23   | root | 39 | 19 | 0       | 0     | 0     | R | 0.7  | 0.0  | 0:00.83  | khugepaged |
| 53   | root | 20 | 0  | 0       | 0     | 0     | S | 0.7  | 0.0  | 0:00.02  | kswapd0    |
| 1    | root | 20 | 0  | 179204  | 13400 | 8412  | S | 0.3  | 0.3  | 0:08.87  | systemd    |
| 903  | root | 20 | 0  | 235036  | 3596  | 3216  | S | 0.3  | 0.1  | 0:00.89  | bash       |

```

[ 1484.694715] Node 0 DMA32: 6*4kB (UME) 18*8kB (UME) 4*16kB (E) 0*32kB (UE) 6*64kB (UME) 5*128kB (UME) 2*256kB (UE) 3*512kB (UME) 1*1024kB (UE) 0*2048kB (UE) 0*4096kB (M) = 39400kB
[ 1484.699307] Node 0 Normal: 164*4kB (UMEH) 296*8kB (UMEH) 144*16kB (UEH) 90*32kB (UEH) 35*64kB (UEH) 16*128kB (UEH) 3*256kB (UEH) 2*512kB (UEH) 0*1024kB (UEH) 0*2048kB (M) 1*4096kB (M) = 35792kB
[ 1484.704255] Node 0 hugepages_total=0 hugepages_free=0 hugepages_surp=0 hugepages_size=2048kB
[ 1484.706703] 9699 total pagecache pages
[ 1484.709295] 9669 pages in swap cache
[ 1484.711797] Swap cache stats: add 2793256, delete 2783587, find 1337968/1506795
[ 1484.714354] Free swap = 0kB
[ 1484.716897] Total swap = 839676kB
[ 1484.719476] 1040444 pages RAM
[ 1484.721988] 0 pages HighMem/MovableOnly
[ 1484.724506] 81173 pages reserved
[ 1484.727028] 0 pages huPoisoned
[ 1484.729572] [ pid ] uid tgid total_vm rss pgtables_bytes swapents oom_score_adj name
[ 1484.732207] [ 666 ] 0 666 23507 15 303104 771 0 systemd-journal
[ 1484.734837] [ 692 ] 0 692 27300 12 221184 806 -1000 systemd-udevd
[ 1484.737467] [ 800 ] 32 800 16700 6 172032 174 0 rpcbind
[ 1484.740093] [ 811 ] 0 811 16055 7 135168 156 -1000 auditd
[ 1484.742730] [ 839 ] 998 839 406405 0 339968 1787 0 polkitd
[ 1484.745379] [ 841 ] 0 841 115681 0 413696 465 0 ModemManager
[ 1484.748007] [ 842 ] 0 842 4223 6 65536 31 0 mcelog
[ 1484.750748] [ 843 ] 81 843 22318 4 167936 215 -900 dbus-daemon
[ 1484.753407] [ 844 ] 0 844 6612 7 94208 298 0 smartd
[ 1484.756066] [ 845 ] 0 845 105552 2 442368 491 0 sssd
[ 1484.758713] [ 849 ] 997 849 4188 7 69632 29 0 lsmd
[ 1484.761369] [ 853 ] 70 853 20655 7 172032 113 0 avahi-daemon
[ 1484.764017] [ 857 ] 0 857 77806 0 104320 169 0 gssproxy
[ 1484.766654] [ 866 ] 991 866 21624 0 204800 190 0 rngd
[ 1484.769300] [ 867 ] 70 867 20655 0 151552 100 0 avahi-daemon
[ 1484.771937] [ 873 ] 0 873 107187 18 434176 614 0 sssd_be
[ 1484.774560] [ 875 ] 0 875 107745 19 442368 307 0 sssd_nss
[ 1484.777179] [ 877 ] 0 877 124600 5 479232 5450 0 firewalld
[ 1484.779849] [ 884 ] 0 884 24456 7 217008 652 0 systemd-logind
[ 1484.782470] [ 893 ] 0 893 150492 3 397312 1054 0 NetworkManager
[ 1484.785046] [ 903 ] 0 903 50759 30 81920 57 0 bash
[ 1484.787566] [ 907 ] 0 907 23242 0 212992 229 -1000 sshd
[ 1484.789980] [ 910 ] 0 910 150766 131 450560 3573 0 tuned
[ 1484.792360] [ 917 ] 0 917 50759 0 69632 95 0 bash
[ 1484.794664] [ 918 ] 0 918 54270 0 65536 23 0 tail
[ 1484.796883] [ 1035 ] 0 1035 61464 0 110592 224 0 crond
[ 1484.799041] [ 1040 ] 0 1040 30419 0 266240 225 0 login
[ 1484.801127] [ 1046 ] 0 1046 10656 4 114608 47 0 atd
[ 1484.803214] [ 1345 ] 0 1345 23434 2 212992 354 0 systemd
[ 1484.805247] [ 1350 ] 0 1350 37935 0 290016 1300 0 (sd-pan)
[ 1484.807224] [ 1356 ] 0 1356 59116 1 94208 407 0 bash
[ 1484.809123] [ 1452 ] 0 1452 1146305 903040 8014592 106009 0 mem.bash
[ 1484.811016] [ 1453 ] 0 1453 60572 106 163040 90 0 top
[ 1484.813043] Out of memory: Killed process 1452 (mem.bash) total-vm:4505220kB, anon-rss:3615360kB, file-rss:0kB, shmem-rss:0kB, UID:0

```

```

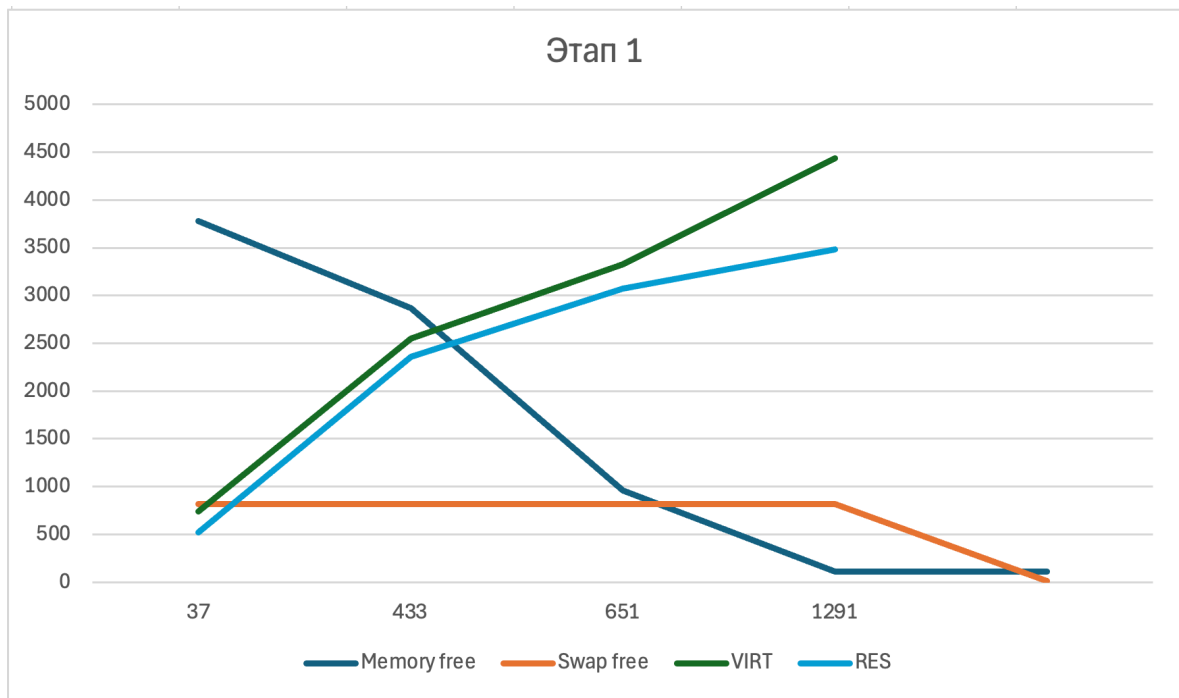
[ 1486.494813] Out of memory: Killed process 1452 (mem.bash) total-vm:4505220kB, anon-rss:3615360kB, file-rss:0kB, shmem-rss:0kB, UID:0
[ 1486.494813] oom_reaper: reaped process 1452 (mem.bash), now anon-rss:0kB, file-rss:0kB, shmem-rss:0kB

```

```

[root@localhost lab5]# tail -1 report.log
55000000
[root@localhost lab5]#

```



## 2. Этап 2

```
#!/bin/bash

: > report2.log

arr=()
stop=0

while true
do
    arr+=(1 2 3 4 5 6 7 8 9 10)
    ((step++))

    if ((step % 100000 == 0)); then
        echo "${#arr[@]}" >> report2.log
    fi
done
```

```
#!/bin/bash

./mem.bash &
./mem2.bash &
```

```

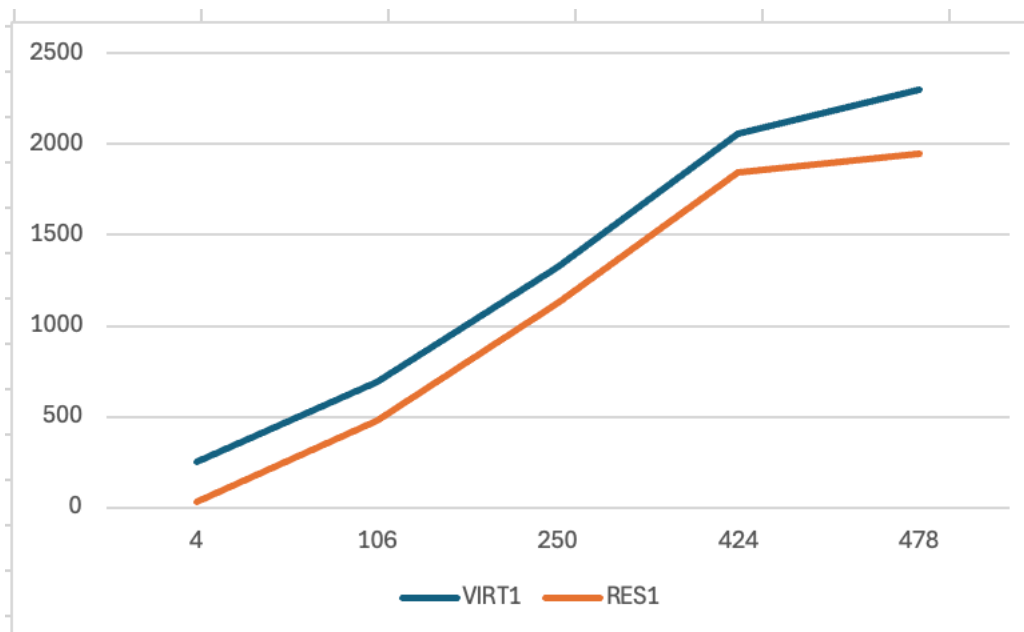
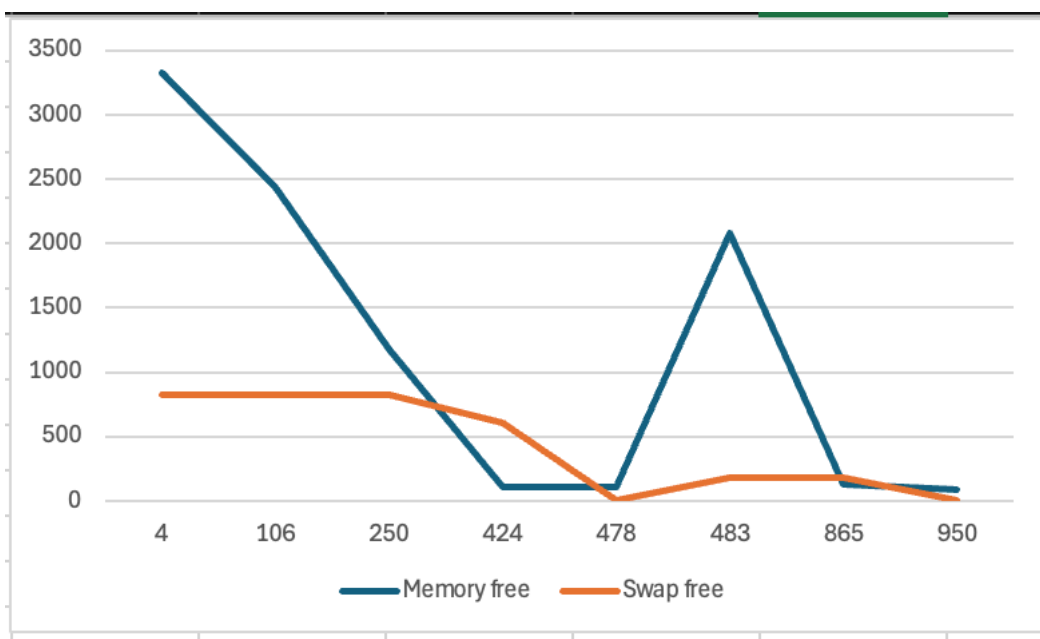
[root@localhost lab5]# dmesg | grep "mem.bash" | tail -2
[ 618.437831] [ 1443]   0 1443  590453 488991 4362240  45887          0 mem.bash
[ 618.443298] Out of memory: Killed process 1443 (mem.bash) total-vm:2361812kB, anon-rss:1955964kB, file-rss:0kB, shmem-rss:0kB, UID:0
[root@localhost lab5]# dmesg | grep "mem2.bash" | tail -2
[ 900.646560] Out of memory: Killed process 1444 (mem2.bash) total-vm:4612544kB, anon-rss:3634492kB, file-rss:0kB, shmem-rss:0kB, UID:0
[ 902.085030] oom_reaper: reaped process 1444 (mem2.bash), now anon-rss:0kB, file-rss:0kB, shmem-rss:0kB

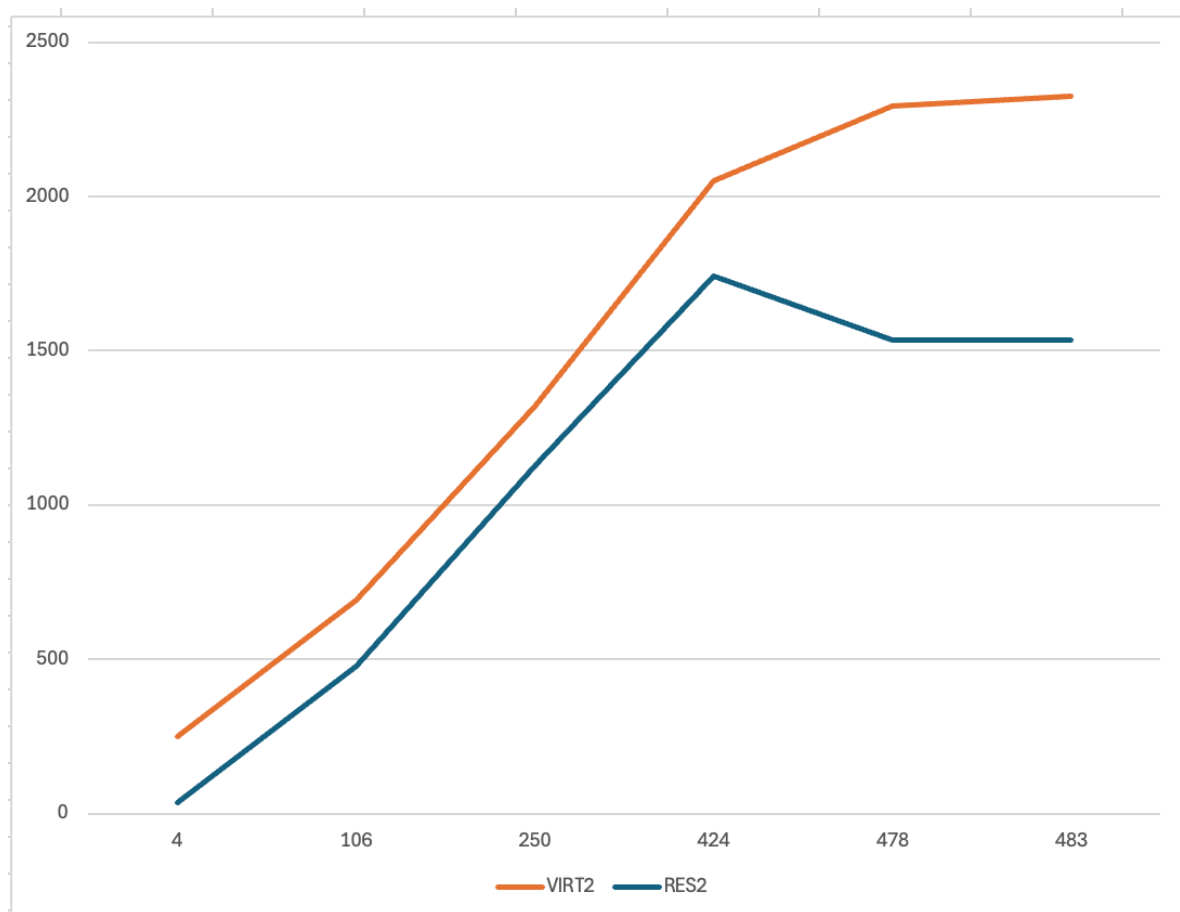
```

```

[root@localhost lab5]# tail -1 report.log
27000000
[root@localhost lab5]# tail -1 report2.log
56000000
[root@localhost lab5]# _

```





**Вывод:** В эксперименте было показано, что при увеличении потребления памяти процессом сначала используется оперативная память, затем подключается swar. При полном исчерпании ресурсов система завершает процесс с помощью OOM Killer, что подтверждается записями в dmesg.

При запуске двух процессов память заканчивается быстрее, и один из процессов завершается раньше из-за конкуренции за ресурсы.

## Эксперимент №2

```
[root@localhost lab5]# ./run_newmem.bash 5500000 10
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
newmem.bash done
All processes finished
```

Попробовал 5500000

```
[ 1129.656864] Out of memory: Killed process 1452 (neumen.bash) total-vm:634328kB, anon-rss:335072kB, file-rss:0kB, shmem-rss:0kB, UID:0
neumen.bash done
neumen.bash done
neumen.bash done
neumen.bash done
./run_neumen.bash: line 12: 1438 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1440 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1442 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1444 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1450 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1452 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1454 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1458 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1460 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1464 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1466 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1468 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1470 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1472 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1476 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1478 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1482 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1484 Killed                  ./neumen.bash "$N"
./run_neumen.bash: line 12: 1486 Killed                  ./neumen.bash "$N"
neumen.bash done
neumen.bash done
neumen.bash done
neumen.bash done
neumen.bash done
neumen.bash done
All processes finished
root@localhost lab51#
```

Попробовал 2000000



Попробовал 2200000

[illegible]

Попробовал 2250000



